## Element C: Presentation and Justification of Solution Design Requirements

Power cables often come unplugged accidentally for many reasons, like getting kicked or pulled from their socket unintentionally. This affects the general public. The goal of our project is to prevent cables from becoming unplugged by accidentally getting kicked or pulled on. This will help people prevent loss of computer data, loss of aborted creations when 3D printing, or loss of electricity to refrigerators and freezers.

Prioritized list of design requirements:

1. Safe to use- (high)
2. Doesn’t permanently damage outlets or plugs-(high)
3. Has to be less than $0.50 per outlet or cable secured-(High)
4. Durable enough for at least five installments-(High)
5. Can be used on a large variety of outlets and cables-(high)
6. Has to be easily installed in under 60 seconds-(medium)

Our product needs to be made of a non-conductive material, as it is used with outlets. It cannot permanently damage or alter the outlet or cable in any way. It has to be less than one dollar per outlet, or else it will be more expensive than the current available solutions. It needs to be durable enough to be installed and uninstalled at least five times before it stops working. It needs to be compatible with a wide variety of outlets and cables. This is especially important because people do not want to have to buy different versions of our product if they have a variety of different power strips and outlets in their house.

Our medium priority design requirements are: it needs to be installed in under 60 seconds because people are more likely to use it if installation is easy. It has to be easy enough to install for most people. These requirements will result in a solution that is useful for the stakeholder while meeting all of their needs. These will meet the requirements of Safe to use-High, Doesn’t damage outlets, Has to be less than $1 per outlet or cable secured, durable enough for at least five installments, has to be easily implemented, and can be installed in under 60 seconds.

Mr. L, who is a stakeholder because he works from home on a computer and uses cables all day. Mr. L agrees with all the design requirements. He said that safety was very important. We are working with high voltage. He wouldn't pay more than five dollars for a solution. He also said the ability to be restuck was important because he would not want it attached forever, and the versatility was not as important because power cables have a standard size.